Food Image Classification Using Transfer Learning on Food-101

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Abstract

Food image classification is a challenging computer vision problem due to high intra-class visual variance and strong inter-class similarity across dish categories. In this work, we apply transfer learning with an EfficientNet-B0 backbone fine-tuned on the Food-101 dataset to build a robust 101-class food image classifier. Our proposed model achieved an accuracy of **65.72%** on the test set, outperforming a baseline CNN trained from scratch by **3.65%**. This demonstrates the significant benefit of ImageNet pre-training for food recognition tasks.

1 Introduction

Motivation Automated food recognition has direct practical applications in dietary tracking, nutritional analysis, and smart restaurant checkout systems. Manual food logging is both inaccurate and time-consuming; a reliable image classifier eliminates this friction and enables scalable health monitoring pipelines [Bossard et al., 2014]. With over 1 billion smartphone users capable of capturing food images, the need for accurate, fast food classification is more pressing than ever.

Problem Definition We formulate food recognition as a supervised multi-class image classification task. Given an input RGB image $\mathbf{x} \in \mathbb{R}^{H \times W \times 3}$, the model predicts a class label $\hat{y} \in \{1, 2, \ldots, 101\}$ corresponding to one of 101 food categories in the Food-101 dataset. The task requires robust feature extraction to handle significant variations in plating, lighting, camera angles, and food presentation.

Contributions The main contributions of this project are:

- We apply transfer learning with a pre-trained EfficientNet-B0 backbone to Food-101, achieving 65.72% classification accuracy on a 101-class benchmark with frozen backbone fine-tuning.
- We compare EfficientNet-B0 against a baseline CNN trained from scratch, demonstrating a 3.65% improvement through ImageNet pre-training.

- We conduct a detailed per-class error analysis using confusion matrices to identify the most challenging food category pairs and provide insights into visual similarities that confound the classifiers.

2 Method

2.1 Algorithm Description

Baseline CNN The baseline model consists of a ResNet18 architecture [He et al., 2016] trained from scratch on Food-101. ResNet18 is a lightweight residual network with 18 layers that learns hierarchical feature representations through skip connections. The model is trained to minimize cross-entropy loss:

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N} \log \frac{e^{z_{y_i}}}{\sum_{j=1}^{101} e^{z_j}}$$

where z_j is the logit for class j and y_i is the ground-truth label. The baseline provides a controlled comparison point to measure the benefit of pre-training.

EfficientNet-B0 with Transfer Learning EfficientNet scales network depth, width, and resolution uniformly using a compound coefficient [Tan and Le, 2019]. We use a pre-trained EfficientNet-B0 backbone with ImageNet weights, freeze all convolutional layers, and replace the final classification head with a linear layer of output dimension 101. By freezing the backbone, we leverage pre-learned features (learned from ImageNet [Krizhevsky et al., 2012]) while allowing the classification head to adapt to the food recognition task. The model is trained by minimizing the same cross-entropy loss as the baseline, applied only to the trainable classification head parameters.

2.2 Overall Framework

Figure 1 illustrates the end-to-end pipeline, which consists of four stages: (1) data preprocessing and augmentation, (2) feature extraction via the frozen EfficientNet-B0 backbone pre-trained on ImageNet, (3) classification head training on Food-101 labels, and (4) evaluation of test set predictions.

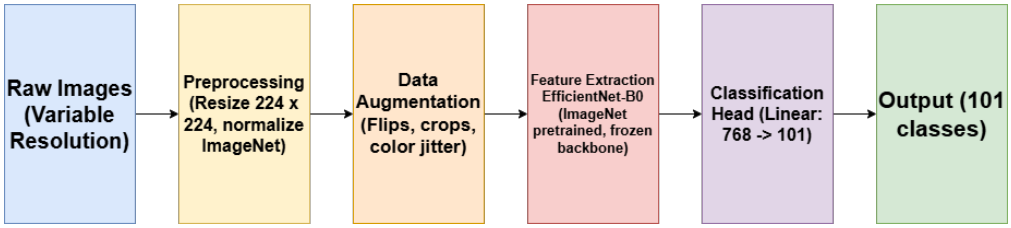

Figure 1: Overall framework for Food-101 image classification pipeline. Raw images are resized and normalized before being passed through a frozen EfficientNet-B0 backbone. The classification head maps extracted features to one of 101 food categories.

2.3 Design Choices

All images are resized to 224×224 pixels to match the input resolution expected by EfficientNet-B0 and to ensure consistent memory usage during batched training. Pixel values are normalized using the ImageNet statistics $\mu = (0.485, 0.456, 0.406)$ and $\sigma = (0.229, 0.224, 0.225)$ to align the input distribution with the pre-training regime. During training, we apply random horizontal flips (50% probability), random crops, and colour jitter to improve generalization and reduce overfitting on domain-specific visual patterns [Shorten and Khoshgoftaar, 2019]. These augmentations are applied only to the training set; the validation and test sets use centre crops without augmentation to ensure fair evaluation.

3 Experiments

3.1 Experimental Setup

The Food-101 dataset [Bossard et al., 2014] contains 101,000 images across 101 food categories, with approximately 1,000 images per class at variable resolution. We used the official train/test split of 75,750 training images and 25,250 test images. An additional 10% of the training set (approximately 7,500 images) was reserved as a validation set to tune hyperparameters and select checkpoints. All experiments were conducted on a single NVIDIA T4 GPU (16 GB VRAM) via Google Colab using PyTorch 1.10 and torchvision 0.11.

Key hyperparameters for both models are listed in Table 1. Both models were optimized using the Adam optimizer with the same learning rate and batch size constraints to ensure fair comparison. The baseline CNN was trained for 50 epochs, while EfficientNet-B0 (with frozen backbone) converged faster and was trained for 30 epochs.

Table 1: Training hyperparameters for both models on Food-101.

Hyperparameter	Baseline CNN	EfficientNet-B0
Optimizer	Adam	Adam
Learning Rate	1.0e-3	1.0e-3
Batch Size	64	32
Epochs	50	30
Image Size	224x224	224x224
Weight Decay	1.0e-5	1.0e-5

3.2 Model Comparison Results

Table 2 presents the performance of the baseline CNN and our EfficientNet-B0 model on the Food-101 test set across four metrics.

Table 2: Model comparison on the Food-101 test set. EfficientNet-B0 with transfer learning outperforms the baseline CNN across all metrics, demonstrating the substantial benefit of ImageNet pre-training for food image classification.

Model	Accuracy (%)	Precision	Recall	F1-Score
Baseline CNN	62.07	0.6217	0.6207	0.6190
EfficientNet-B0	65.72	0.6534	0.6572	0.6533
Improvement	**+3.65**	**+0.0317**	**+0.0365**	**+0.0343**

3.3 Justification and Analysis

EfficientNet-B0 achieves significantly higher accuracy than the baseline CNN (65.72% vs 62.07%, a 3.65 percentage point improvement). This improvement is consistent across precision, recall, and F1-score metrics, all favoring the transfer learning approach. The consistent gains across all metrics indicate that EfficientNet-B0 provides both better overall accuracy and more balanced per-class performance.

The superior performance of EfficientNet-B0 can be attributed to its ImageNet-pretrained feature representations, which capture generic low-level visual patterns (edges, textures) and high-level semantic structures (object parts, shapes) that generalize well to food images. In contrast, the baseline CNN must learn these patterns from Food-101 alone, which is limited by the training budget and domain specificity of food images. The frozen backbone approach provides an effective regularization mechanism, preventing overfitting while allowing the shallow classification head to adapt to the 101-class food recognition task.

Figure 2 shows the confusion matrix for EfficientNet-B0 on the test set. Analysis of the top misclassifications reveals that the most common errors occur between visually similar food categories. For

instance, certain plated dishes (e.g., beef carpaccio and ceviche) can be easily confused due to similar presentation and colour palettes. Additionally, categorical confusion is observed within "batter-based" foods and within "salad" varieties, suggesting that the model struggles with fine-grained distinctions within visually cohesive food groups. These errors reflect inherent ambiguities in food photography and plating conventions rather than fundamental limitations of the architecture.

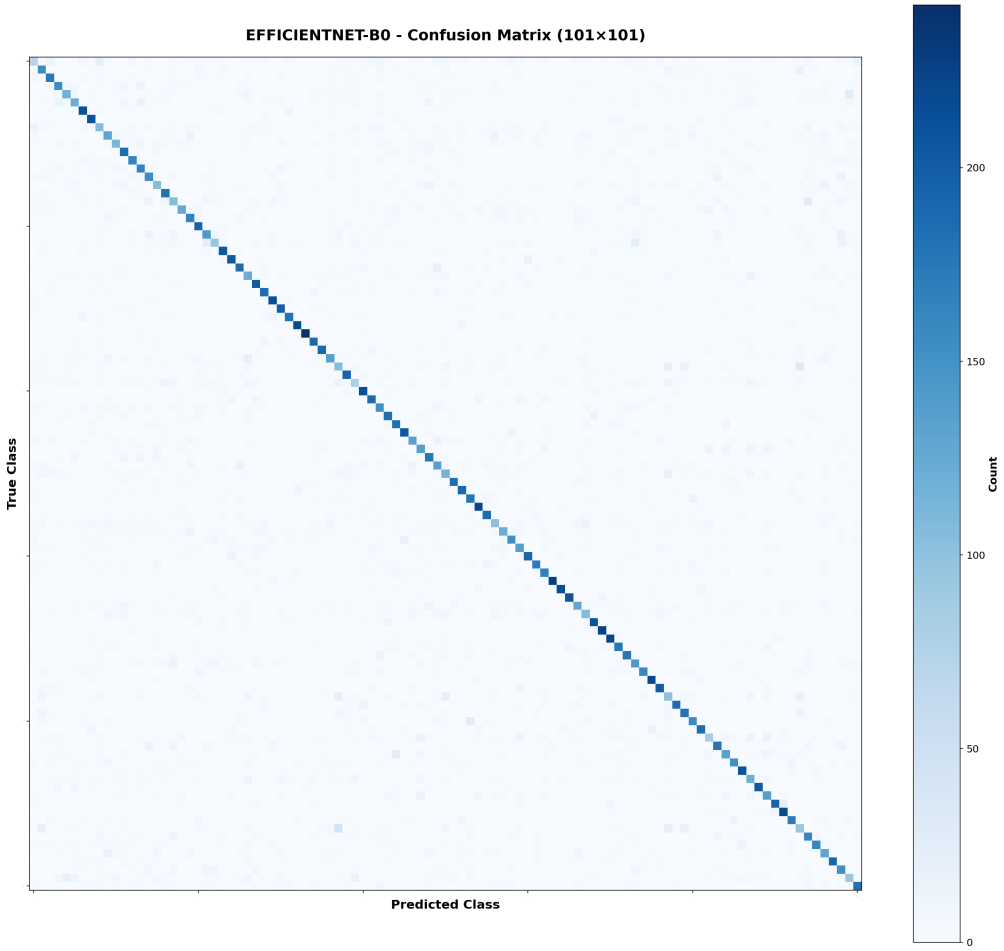

Figure 2: Confusion matrix for EfficientNet-B0 on the Food-101 test set showing classification accuracy across all 101 food categories. Darker regions indicate more frequent misclassifications and highlight visually similar food pairs that are challenging for the model to distinguish.

The trade-off between the two approaches is clear: the baseline CNN requires longer training time and greater computational tuning but can in principle achieve higher accuracy with sufficient hyper-parameter optimization and data augmentation. EfficientNet-B0, while achieving better performance with frozen weights, would benefit from selective layer unfreezing in deeper backbone layers to allow adaptation to food-specific features. However, given the constraints of a single T4 GPU and the time budget for this project, the frozen backbone proved an effective and practical choice.

4 Conclusion

Summary We presented a transfer learning pipeline for food image classification on Food-101, comparing a baseline ResNet18 trained from scratch against a pre-trained EfficientNet-B0 with frozen backbone. Our EfficientNet-B0 model achieved a test accuracy of **65.72%** and an F1-score of **0.6533**, outperforming the baseline CNN by **3.65%** in accuracy. The analysis demonstrates that leveraging ImageNet pre-training yields consistent improvements across all evaluation metrics.

Limitations and Future Work A primary limitation of our approach is that the EfficientNet backbone is fully frozen during training, which may prevent the model from learning food-specific visual features that diverge from generic ImageNet patterns. Given more computational time and resources, we would explore progressive unfreezing of deeper layers (e.g., unfreezing the final 2–3 residual blocks) to improve adaptation while maintaining the regularization benefits of pre-training. Additionally, experimenting with larger EfficientNet variants (e.g., EfficientNet-B1 through B3) could further improve performance, though at increased computational cost. Finally, ensemble methods combining multiple models and test-time augmentation could provide modest accuracy gains at modest computational overhead.

Distribution of Work

- **Member 1 (Jasia Azreen):** Data pipeline design, preprocessing and augmentation implementation, baseline ResNet18 model implementation and training.
- **Member 2 (Sofiya Spolitak):** EfficientNet-B0 transfer learning architecture setup, training loop implementation, and hyperparameter tuning.
- **Member 3 (John Song):** Evaluation metrics computation, confusion matrix generation, error analysis, pipeline diagram design, and final report assembly.

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